JAY M. SEN

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Education

Rensselaer Polytechnic Institute (RPI) | Rensselaer Medalist | Troy, NY Expected December 2028

Bachelor of Science in Mechanical Engineering, *3.5 GPA*

Relevant Coursework: Solid Mechanics, Numerical Programming, Thermodynamics, Statics, Material Science, Engineering Processes, Calculus I-III, Differential Equations, Linear Algebra, Physics I-II, CAD, Applied Statistics, Chemistry I-II

Work Experience

Production Engineering Intern | Cleaveland/Price Inc. | Trafford, PA May 2026 - August 2026

- Conducted 180+ time studies on machining and assembly operations, capturing cycle times, setup times, and process variables
- Developed 12+ templates and 2 AI agents to standardize the time study process, improving data consistency and repeatability
- Built the data foundation for job time calculators, new facility planning, and equipment layout decisions
- Built 2 calculators to estimate job times based on part characteristics and process parameters
- Documented sources of wasted time and interviewed 30+ operators to turn observations into improvement recommendations

Director of IT | Sen Engineering Solutions, Inc. | Monroeville, PA June 2024 - Present

- Participated in contract and planning discussions with manufacturers and customers
- Built and launched a fully dynamic Next.js website to repair site functionality and modernize brand identity
- Diagnosed and resolved hardware and software issues regarding computers, security systems, and online meetings

Dining Room Server | Juniper Communities | Monroeville, PA February 2024 - August 2025

- Delivered meals to 60+ residents promptly and accurately according to dietary restrictions and resident preferences
- Fostered a compassionate, positive, and respectful environment for residents during meal times
- Maintained a clean and organized dining room following health and safety protocols
- Communicated regularly with nursing staff to address resident needs or concerns
- Conducted table setup, food prep, and kitchen support

Extracurricular Activities

Rensselaer Motorsport (FSAE) | Troy, NY August 2025 - Present

- Leading a 5 person aerodynamics sub team working to optimize a multi-element wing design
- Machined chassis jigs and fabricated suspension components and carbon fiber aero components
- Conducted Finite Element Analysis on rear wing truss using SolidWorks Simulation, improving Factor of Safety by 1.4
- Served as race marshal, monitoring track conditions and clearing debris to protect drivers and team members

Eagle Scout | Troop 220 | Scouting America September 2013 - January 2026

- Eagle Scout Service Project: planned and built a 10 x 3 ft trail bridge in Boyce Park, Monroeville, PA
- Leadership: Senior Patrol Leader (Feb 2023 - Jun 2025), Patrol Leader (Feb 2022 - Feb 2023)
- Community Service (50+ hours): Community cleanups, trail maintenance, and summer camp maintenance
- Led my troop to 3rd place in the 2024 Lackawanna District Klondike Derby

Cello Performance January 2011 - Present

- 14+ years of private lessons, group lessons, and performances with Rana's Music Studio
- Rensselaer Orchestra
- Pennsylvania Music Educators Association Region 1 finalist (2019)
- Gateway High School Orchestra Performance at Disney Springs, Lake Buena Vista, FL - Principal Cellist (2022)
- Pittsburgh Symphony Orchestra Side-by-Side alternate cellist (2024)

Skills

Software: Python, Siemens NX, SolidWorks, SolidWorks Simulation, Ansys Mechanical, Star-CCM+, Simcenter HEEDS

Manufacturing: Geometric Dimensioning & Tolerancing, Mastercam, Manual Machining, Welding, Lean Manufacturing

Business: Product Development & Launch, Customer Service, Google & Microsoft Suite